Worksheet: work out True or False before you run it

NAME _____ DATE Mon Oct 5

CHUNK 4 OF 6

and: both must be True

`and` is True only when both sides are True.

```
height = 52
has_ticket = True
if height > 48 and has_ticket:
    print("Board the coaster.")
else:
    print("Step aside.")
```

Each row is one rider. Fill in the blanks. Shaded rows are done for you.

rider	height	has_ticket	height > 48?	height > 48 and has_ticket?	what it prints
1	52	True	True	True	Board the coaster.
2	52	False			
3	48	False			
4	48	True	False	False	Step aside.
5	40	True			
6	60	False			
7	49	False	True	False	Step aside.
8	49	True			

Which riders board the coaster? Why?

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or: one is enough

`or` is True when either side is True.

```
is_staff = False
has_pass = True
if is_staff or has_pass:
    print("Gate opens.")
else:
    print("Gate stays shut.")
```

Each row is one visitor. Fill in the blanks. The shaded row is done for you.

visitor	is_staff	has_pass	is_staff or has_pass?	what it prints
1	False	True	True	Gate opens.
2	True	False		
3	True	True		
4	False	False		

Which visitor is kept out? Why?

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A while loop and its stopping condition

`!=` means is not equal to. `""` is empty text.

```
answer = ""
while answer != "yes" and answer != "no":
    answer = input("Ready? Type yes or no: ")
print("Recorded:", answer)
```

Each row is one check of the condition. Fill in the blanks. The shaded row is done for you.

check	answer	answer != "yes"?	answer != "no"?	answer != "yes" and answer != "no"?
1	""	True	True	True
2	"maybe"			
3	"soon"			
4	"yes"			

How many times is the question asked? Which check stops the loop?

The same loop with or

Line 2 now says `or`. Trace it on paper only.

```
answer = ""
while answer != "yes" or answer != "no":
    answer = input("Ready? Type yes or no: ")
print("Recorded:", answer)
```

check	answer	answer != "yes"?	answer != "no"?	answer != "yes" or answer != "no"?
1	""	True	True	True
2	"maybe"			
3	"yes"			
4	"no"			

Does this loop ever stop? Why?